

Experiment 3

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as `ps`, `top`, and `/proc`. Document the observations.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    pid_t pid;
```

```
    printf("Before fork\n");
```

```
    printf("Process PID: %d\n", getpid());
```

```
    pid = fork();
```

```
    if (pid < 0)
```

```
{
```

```
perror("fork failed");  
return 1;  
}  
  
if (pid == 0)  
{  
    // Child process  
    printf("\nChild Process\n");  
    printf("PID : %d\n", getpid());  
    printf("PPID : %d\n", getppid());  
    printf("Child process is running...\n");  
  
    // Keep child alive for observation  
    sleep(20);  
  
    printf("Child process is terminating...\n");  
    return 0;  
}  
else  
{  
    // Parent process  
    printf("\nParent Process\n");  
    printf("PID : %d\n", getpid());
```

```
printf("PPID : %d\n", getppid());  
printf("Child PID: %d\n", pid);  
  
printf("Parent process is waiting...\n");  
  
// Keep parent waiting while child runs  
wait(NULL);  
  
printf("Child process has terminated.\n");  
printf("Parent process is terminating...\n");  
}  
  
return 0;  
}
```

Output:

```
ashwithreddy_earla@ASHWITHREDDYPC:~$ gcc experiment3.c -o experiment3
ashwithreddy_earla@ASHWITHREDDYPC:~$ ./experiment3
```

Parent process started.

Parent PID: 1104

Parent PPID: 912

Parent process.

Parent PID: 1104

Child PID: 1105

Parent is waiting for child...

Child process created.

Child PID: 1105

Child PPID: 1104

Child process is running...

Child process is terminating.

Child process has terminated.

Parent process is terminating.

```
ashwithreddy_earla@ASHWITHREDDYPC:~$ ps -ef
UID          PID     PPID  C  STIME TTY          TIME CMD
root           1         0  0  03:02 ?        00:00:00 /sbin/init
root           2         1  0  03:02 hvc0     00:00:00 /init
root           7         2  0  03:02 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          47         1  0  03:02 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+      79         1  0  03:02 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          86         1  0  03:02 ?        00:00:01 /usr/lib/systemd/systemd-udevd
root         108         1  0  03:02 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         109         1  0  03:02 ?        00:00:00 /usr/sbin/cron -f -P
message+     110         1  0  03:02 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-
root         116         1  0  03:02 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         140         1  0  03:02 ?        00:00:00 /usr/lib/systemd/systemd-logind
syslog       170         1  0  03:02 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         196         1  0  03:02 tty1     00:00:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/
_chrony      223        108  0  03:02 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
_chrony      229        223  0  03:02 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         237         1  0  03:02 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signa
root         311         2  0  03:02 ?        00:00:00 /init
root         312        311  0  03:02 ?        00:00:00 /init
ashwith+     313        312  0  03:02 pts/0    00:00:00 -bash
root         314         2  0  03:02 ?        00:00:00 login -- ashwithreddy_earla
ashwith+     358         1  0  03:02 ?        00:00:00 /usr/lib/systemd/systemd --user
ashwith+     361        358  0  03:02 ?        00:00:00 (sd-pam)
ashwith+     387        314  0  03:02 pts/1    00:00:00 -bash
root         479         2  0  03:03 ?        00:00:00 /init
root         480         2  0  03:03 ?        00:00:00 /init
ashwith+     486        480  0  03:03 pts/2    00:00:00 -bash
root         905         2  0  03:32 ?        00:00:00 /init
root         906        905  0  03:32 ?        00:00:00 /init
ashwith+     912        906  0  03:32 pts/3    00:00:00 -bash
ashwith+    1105        912  0  03:52 pts/2    00:00:00 ps -ef
```